Chapter 1 Study Guide: Vector Spaces

Sections 1.1 – 1.5

Based on Nair & Singh, *Linear Algebra* (Springer)

Intuition

Everything in this chapter answers one question: **what is the minimum set of rules that "vector-like" behavior must follow, so all the algebra we're used to (adding things, scaling things) still works — no matter what the "things" actually are?** Once you accept that vectors don't have to be arrows, but can be polynomials, matrices, functions, or sequences, the whole chapter becomes a five-step ladder:

1. **1.1** Define the rules (vector space axioms).
2. **1.2** Find smaller vector spaces living inside a bigger one (subspaces).
3. **1.3** Build a subspace by combining vectors (span).
4. **1.4** Ask whether a set of vectors has redundancy (linear independence).
5. **1.5** Find the most efficient spanning set (basis), and count its size (dimension).

Each rung needs the one below it — keep this ladder in mind throughout.

1 Vector Space

1.1 Start with something familiar

Picture an ordinary arrow in the plane, starting at the origin. You can do exactly two things with arrows:

- **add** two arrows (tip-to-tail, or componentwise if both start at the origin),
- **scale** an arrow by a number (stretch, shrink, or flip it).

Nothing else about "arrow-ness" — length, angle, direction — was actually needed to do algebra with them. Mathematicians stripped away the picture and kept only the two operations, together with the algebraic rules they obey. Any set equipped with those two operations, obeying those rules, is called a **vector space** — whether or not its elements look anything like arrows.

1.2 The formal definition

A **vector space over a field** F (in this book, F is always $\mathbb{R}$ or $\mathbb{C}$) is a nonempty set V with two operations:

- **addition**: takes $x, y \in V$, produces $x + y \in V$;
- **scalar multiplication**: takes $\alpha \in F$, $x \in V$, produces $\alpha x \in V$;

satisfying 8 axioms. Don't memorize them as an isolated list — group them by what job each one does:

Group	Axioms	What it says
Addition behaves normally	commutative, associative	$x + y = y + x, \quad (x + y) + z = x + (y + z)$
Addition has an identity and inverses	zero vector, additive inverse	$\exists\, 0 : x + 0 = x; \quad \exists -x : x + (-x) = 0$
Scalars distribute over vectors	two distributive laws	$\alpha(x + y) = \alpha x + \alpha y, \quad (\alpha + \beta)x = \alpha x + \beta x$
Scalar multiplication is consistent	associativity, identity	$(\alpha\beta)x = \alpha(\beta x), \quad 1x = x$

Common Mistake

The symbol "+" secretly means *two different things*: adding two scalars in F, versus adding two vectors in V. The same goes for "0" (zero scalar vs. zero vector) and for writing αx (scalar×scalar vs. scalar×vector). This is not sloppy notation — it's deliberate, and you're expected to infer from context which one is meant. When you get confused mid-proof, stop and ask: *"is this + or 0 living in F or in V right now?"*

Uniqueness (Theorem 1.2). The zero vector is unique, and every vector's additive inverse is unique. This is precisely why we're allowed to say *"the"* zero vector instead of *"a"* zero vector — uniqueness is what earns it one fixed symbol.

1.3 The zoo of examples — where students usually get stuck

The genuinely new idea in 1.1 is that vector spaces are everywhere, not just $\mathbb{R}^2$ and $\mathbb{R}^3$.

Worked Example : F^n, the space you already know

$F^n = \{(a_1, \ldots, a_n) : a_i \in F\}$, with addition and scaling done slot-by-slot:

$$(a_1, \ldots, a_n) + (b_1, \ldots, b_n) = (a_1 + b_1, \ldots, a_n + b_n), \qquad \alpha(a_1, \ldots, a_n) = (\alpha a_1, \ldots, \alpha a_n).$$

This is ordinary $\mathbb{R}^2/\mathbb{R}^3$-style vector algebra, just generalized to n slots.

Worked Example : matrices, $F^{m\times n}$

All $m \times n$ matrices, added entry-by-entry, scaled entry-by-entry. A matrix here is nothing mysterious — it's just "a big vector whose entries happen to be arranged in a grid." For instance in $\mathbb{R}^{2\times 2}$:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 4 & 4 \end{pmatrix}, \qquad 3\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}.$$

Worked Example : polynomials, $P_n(F)$

$P_n(F)$ is all polynomials of degree *at most* n. So $P_2(\mathbb{R})$ contains $3 - t + 5t^2$, and also 7 (degree 0), and also 0. Addition combines like terms; scalar multiplication multiplies every coefficient. E.g., in $P_2(\mathbb{R})$:

$$(1 + 2t - t^2) + (3 - t + 4t^2) = 4 + t + 3t^2, \qquad 2(1 + 2t - t^2) = 2 + 4t - 2t^2.$$

Why "at most n" and not "exactly n"? Because the set of degree-*exactly*-n polynomials is not closed under addition: take t^2 and $1-t^2$, both degree 2 in appearance, but $t^2 + (1-t^2) = 1$ drops to degree 0. Since a vector space must be closed under addition, "exactly degree n" can never be a vector space, but "at most degree n" can (it comfortably absorbs cancellations like this one).

Worked Example : function spaces

Let $C(I, \mathbb{R})$ be all continuous real functions on an interval I. Addition and scaling are done **pointwise**:

$$(x + y)(t) := x(t) + y(t), \qquad (\alpha x)(t) := \alpha\, x(t).$$

This is the exact same idea as componentwise addition in $\mathbb{R}^n$ — just think of a function as "a vector with infinitely many components, one per input value t." If $x(t) = t^2$ and $y(t) = \sin t$, then $(x + y)(t) = t^2 + \sin t$ is a brand-new function, built by adding outputs at every point.

Worked Example : sequences, F^∞

All infinite sequences $(a_1, a_2, a_3, \dots)$ of scalars, added and scaled term-by-term:

$$(1, 2, 3, \dots) + (0, 1, 0, 1, \dots) = (1, 3, 3, \dots).$$

A sequence is just a function from $\mathbb{N}$ into F — another instance of the "function space" idea above.

Common Mistake

$\mathbb{C}$ is a vector space over $\mathbb{C}$ itself, *and* it's also a vector space over $\mathbb{R}$ — these are genuinely different vector spaces even though the underlying points are the same set. "Vector space" is not just a set; it's a set *together with a choice of field*. This distinction resurfaces in Section 1.5, where it changes the dimension you compute.

1.4 Easy-looking facts that still need proof (Theorem 1.6)

In an abstract vector space you cannot lean on intuition about arrows — only the 8 axioms are allowed. That discipline is exactly what these "obvious" facts train:

- $0 \cdot x = 0$ (scalar zero times any vector gives the zero vector)
- $\alpha \cdot 0 = 0$ (any scalar times the zero vector gives the zero vector)
- $(-1)x = -x$
- If $\alpha x = 0$, then $\alpha = 0$ or $x = 0$ (no "zero divisors")

Key Takeaway

A vector space is a set with two operations satisfying 8 rules. The rules are exactly what makes "ordinary algebra" with $+, -, \times$scalar legal. The set of "vectors" can be anything — tuples, matrices, polynomials, functions, sequences — as long as it obeys the rules.

2 Subspaces

2.1 Intuition

A **subspace** is a subset of a vector space that is itself a vector space, using the very same addition and scalar multiplication rules inherited from the parent space. Geometrically in $\mathbb{R}^3$: subspaces are exactly $\{0\}$, lines through the origin, planes through the origin, and all of $\mathbb{R}^3$. Nothing that misses the origin can ever be a subspace.

2.2 The subspace test — your actual day-to-day tool

You almost never re-check all 8 axioms. Instead use the shortcut (Theorem 1.8): a nonempty $U \subseteq V$ is a subspace **if and only if** it's closed under the two operations:

$$x, y \in U,\ \alpha \in F \implies x + y \in U \text{ and } \alpha x \in U.$$

Intuition

Why does closure alone suffice? Because the "harder" axioms (commutativity, associativity, distributivity) hold for *every* element of V, so automatically for elements of U too. The only things that could break are: does U contain 0, and does it contain additive inverses? But closure under scalar multiplication with $\alpha = 0$ gives you $0 \in U$ for free, and $\alpha = -1$ gives you $-x \in U$ for free. That's the slick trick behind the proof.

Practical checklist for "is U a subspace of V?"

1. Is $0 \in U$? (Fast way to rule things out — if $0 \notin U$, stop, it's not a subspace.)
2. Take *arbitrary* $x, y \in U$ and $\alpha \in F$. Is $x + y \in U$? Is $\alpha x \in U$?

Worked Example : a plane through the origin

Is $U = \{(a, b, c) \in \mathbb{R}^3 : a + b + c = 0\}$ a subspace of $\mathbb{R}^3$?

- **Zero:** $(0, 0, 0)$ satisfies $0 + 0 + 0 = 0$. ✓
- **Closed under +:** let $x = (a_1, b_1, c_1)$, $y = (a_2, b_2, c_2) \in U$, so $a_1 + b_1 + c_1 = 0$ and $a_2 + b_2 + c_2 = 0$. Then

$$x + y = (a_1{+}a_2,\ b_1{+}b_2,\ c_1{+}c_2), \quad (a_1{+}a_2) + (b_1{+}b_2) + (c_1{+}c_2) = 0 + 0 = 0. ✓$$

- **Closed under scaling:** $\alpha x = (\alpha a_1, \alpha b_1, \alpha c_1)$, and $\alpha a_1 + \alpha b_1 + \alpha c_1 = \alpha(a_1 + b_1 + c_1) = \alpha \cdot 0 = 0$. ✓

So U is a subspace — geometrically, a plane through the origin, matching intuition.
Contrast: $W = \{(a, b, c) : a + b + c = 1\}$ is *not* a subspace — fails instantly since $0 \notin W$. A plane that misses the origin can never be a subspace, no matter what else is true about it.

2.3 Intersections behave, unions don't

- **Intersection of subspaces is always a subspace** (Theorem 1.10). If both defining conditions are "closed under + and scaling," then requiring *both simultaneously* is still closed.
- **Union of subspaces is usually *not* a subspace.**

Common Mistake

In $\mathbb{R}^2$, let U be the line $y = x$ and W the line $y = 2x$. Take $(1, 1) \in U$ and $(1, 2) \in W$. Their sum $(2, 3)$ lies on neither line, so $U \cup W$ is not closed under addition — it's not a subspace, even though U and W individually are.

Because union fails, we instead define the **sum** of subspaces:

$$U + W := \{u + w : u \in U,\ w \in W\}.$$

This *is* always a subspace (Theorem 1.12) — in fact it's the smallest subspace containing both U and W, i.e. $U + W = \operatorname{span}(U \cup W)$ (this connects forward to Section 1.3).

Worked Example : sum of two planes in $\mathbb{R}^3$

Let $V_1 = \{(a, b, c) : a + b + c = 0\}$ and $V_2 = \{(a, b, c) : a + 2b + 3c = 0\}$, two distinct planes through the origin. Their intersection solves both equations at once: $b = -2a,\ c = a$, giving the line $V_1 \cap V_2 = \{(a, -2a, a) : a \in \mathbb{R}\}$. Meanwhile $V_1 + V_2$ turns out to be *all* of $\mathbb{R}^3$ — two distinct planes through the origin, added together, fill up the whole space. This matches the geometric picture: two lines through the origin sum to (at least) a plane; two planes through the origin sum to (often) the whole space.

Key Takeaway

Subspace = subset closed under + and scalar multiplication (equivalently, containing 0 and stable under those operations). Intersections of subspaces are always subspaces; unions almost never are — use the *sum* $U + W$ instead when you want to combine two subspaces into a bigger one.

3 Linear Span

3.1 Intuition

Given some vectors, what's the *smallest* subspace containing all of them? Answer: take every possible way of scaling and adding them together. That set is the **span**.

3.2 Definitions

A **linear combination** of $u_1, \ldots, u_n$ is any expression $\alpha_1 u_1 + \cdots + \alpha_n u_n$ for scalars α_i. The **span** of a set S, written $\operatorname{span}(S)$, is the set of *all* linear combinations of finitely many vectors from S. By convention $\operatorname{span}(\emptyset) = \{0\}$.

Worked Example : span of two vectors in $\mathbb{R}^3$

Let $S = \{(1,0,0),(0,1,0)\}$. Then

$$\operatorname{span}(S) = \{\alpha(1,0,0) + \beta(0,1,0) : \alpha, \beta \in \mathbb{R}\} = \{(\alpha,\beta,0)\} \;= \text{the entire } xy\text{-plane}.$$

Notice: S itself has only 2 points, but its span is an infinite plane. This is the general pattern — **span turns a small, finite set of vectors into a (usually much bigger) subspace.**

Worked Example : the standard vectors span F^n

Let $e_j \in F^n$ have a 1 in slot j and 0s elsewhere. Then $\operatorname{span}\{e_1, \ldots, e_n\} = F^n$, since any $(\alpha_1, \ldots, \alpha_n) = \alpha_1 e_1 + \cdots + \alpha_n e_n$. This is why they're called *standard* — they're the default building-block basis (formalized in Section 1.5).

Common Mistake

Spanning sets are **never unique**. For instance $\operatorname{span}\{(1,0,0)\} = \operatorname{span}\{(3,0,0)\} = \operatorname{span}\{(\sqrt{2},0,0),(\pi,0,0)\}$ — all describe the same line through the origin. There's no such thing as "*the*" spanning set of a space, only "*a*" spanning set. (Section 1.5 asks: what's the most *efficient* one?)

3.3 Why span matters (Theorem 1.16)

Two equivalent descriptions of $\operatorname{span}(S)$:

1. It's a subspace, and it's the **smallest** subspace containing S (build up from the inside).
2. It equals the **intersection of every subspace that contains** S (carve down from the outside).

These always agree. The standard technique to prove two sets are equal — show each is a subset of the other — appears again and again throughout linear algebra, so it's worth watching closely here.

Key Takeaway

$\mathrm{span}(S)$ = every possible linear combination you can build from S. It is always a subspace, and it's the smallest one containing S. A set "spans" V if $\mathrm{span}(S) = V$.

4 Linear Independence

4.1 Intuition — the crux of the chapter

A spanning set can have **redundant** vectors — vectors that add no new reach because they're already expressible using the others. **Linear independence** formalizes "no redundancy: every vector in this set is pulling its own weight."

4.2 The workhorse test (Theorem 1.26) — what you'll actually use

Vectors $v_1, \ldots, v_n$ are linearly independent **if and only if**

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0 \implies \alpha_1 = \alpha_2 = \cdots = \alpha_n = 0.$$

In words: *the only way to combine them to reach the zero vector is the totally trivial way (all coefficients zero).* If you can find **any** combination with not-all-zero coefficients that still sums to zero, the set is dependent.

Intuition

Recipe, every single time:

1. Write $\alpha_1 v_1 + \cdots + \alpha_n v_n = 0$.
2. Turn this into a system of linear equations in the unknowns α_i (compare components, or coefficients of powers of t, etc.).
3. Solve the system. Only-trivial-solution $\Rightarrow$ independent. Nontrivial solution exists $\Rightarrow$ dependent.

Worked Example : an independent set

Check whether $(1, 0, 0), (1, 1, 0), (1, 1, 1)$ are linearly independent in $\mathbb{R}^3$. Set

$$a(1, 0, 0) + b(1, 1, 0) + c(1, 1, 1) = (0, 0, 0) \implies (a + b + c,\ b + c,\ c) = (0, 0, 0).$$

From the last slot, $c = 0$. Then $b + c = 0 \Rightarrow b = 0$. Then $a + b + c = 0 \Rightarrow a = 0$. Only the trivial solution works, so this set is **linearly independent**.

Worked Example : a dependent set

Now try $(1, 0, 0), (0, 2, 0), (0, 0, 3), (1, 2, 3)$ (four vectors in a 3-dimensional space — already a red flag, see Theorem 1.31 below). Setting $a(1, 0, 0) + b(0, 2, 0) + c(0, 0, 3) + d(1, 2, 3) = 0$ gives

$$(a + d,\ 2b + 2d,\ 3c + 3d) = (0, 0, 0).$$

Choosing $d = 1,\ a = -1,\ b = -1,\ c = -1$ is a **nontrivial** solution (not all zero) that still sums to zero. So this set is **linearly dependent**: indeed $(1, 2, 3) = (1, 0, 0) + (0, 2, 0) + (0, 0, 3)$.

Common Mistake

Linear dependence of a set does **not** mean *every* vector is a combination of the others — it only requires *at least one* to be. E.g., $\{(1, 3, 2), (1, 2, 3), (2, 4, 6)\}$ is dependent because $(2, 4, 6) = 2(1, 2, 3)$, but $(1, 3, 2)$ is *not* a combination of the other two. One "culprit" vector is enough to make the whole set dependent.

Common Mistake

"Independent" does **not** just mean "no two vectors are scalar multiples of each other." That pairwise check only works for exactly 2 vectors. For 3 or more, you must use the full definition (solve $\alpha_1 v_1 + \cdots + \alpha_n v_n = 0$). It's entirely possible for no two vectors in a set to be multiples of one another, yet the whole set is still dependent — e.g., three vectors lying in the same plane in $\mathbb{R}^3$ can point in three visually "different" directions, yet one is still a combination of the other two.

4.3 Quick sanity facts (easy consequences worth memorizing)

- $\emptyset$ is independent (vacuously true — there's nothing to combine).
- $\{0\}$ (just the zero vector alone) is always dependent, since $1 \cdot 0 = 0$ is a nontrivial-looking combination... more directly, $0 \in \text{span}(\emptyset)$.
- Any set containing the zero vector is automatically dependent.
- Subsets of independent sets are independent; supersets of dependent sets are dependent — dependence "spreads upward," independence only "shrinks downward." Think of it as a weakest-link property.

4.4 The counting punchline (Theorem 1.31)

If S is a *finite* spanning set of V with $|S| = n$ vectors, then **any set with more than n vectors must be linearly dependent.**

Intuition

You cannot fit more than n "independent directions" into a space that only needs n vectors to reach everywhere — any extra vector is forced to be redundant. This single theorem is what makes *dimension* well-defined in Section 1.5.

Key Takeaway

To test independence: set a linear combination equal to 0 and solve for the coefficients. All-zero-only solution = independent. If you already have more vectors than the size of a known spanning set, you can conclude dependence immediately, with no computation.

5 Basis and Dimension

5.1 Intuition

Combine 1.3 and 1.4: a **basis** is a spanning set with *zero redundancy* — it spans the whole space, *and* it's linearly independent. It's the "just right" set: not so small that it misses part of the space, not so big that it wastes vectors.

5.2 Definition

$B \subseteq V$ is a **basis** of V if (1) B spans V, and (2) B is linearly independent.

5.3 Why bases matter: unique coordinates (Theorem 1.34)

This is the payoff of the whole chapter. If $\{u_1, \dots, u_n\}$ is an (ordered) basis of V, then **every** vector $x \in V$ can be written as

$$x = \alpha_1 u_1 + \cdots + \alpha_n u_n$$

for a **unique** tuple $(\alpha_1, \dots, \alpha_n)$. These α_i are the **coordinates** of x relative to that basis.

Intuition

This is exactly how (x, y, z)-coordinates work in $\mathbb{R}^3$ relative to the standard basis $\{e_1, e_2, e_3\}$ — except now you know *why* coordinates are well-defined (uniqueness comes straight from linear independence), and you know the idea generalizes: polynomials, matrices, functions — any vector space with a basis gets its own coordinate system.

5.4 Three equivalent characterizations (Theorem 1.37) — internalize this

B is a basis of V **if and only if** any *one* of these holds (all three are equivalent):

1. B spans V and is linearly independent *(the definition)*.
2. B is a **minimal spanning set** — spans V, but no proper subset of it does.
3. B is a **maximal linearly independent set** — independent, but adding any new vector from V breaks independence.

Intuition

Why must these agree? If a spanning set had redundancy, you could shrink it (remove the redundant vector, still spans) — so it wasn't minimal, contradiction. If an independent set weren't maximal, you could add a new vector and stay independent — meaning it hadn't spanned everything yet. A basis sits exactly at the point where shrinking breaks spanning *and* growing breaks independence.

5.5 Dimension is well-defined (Theorem 1.38) — this genuinely needs proof

Question: could a vector space have two different bases, one with 3 vectors and another with 5?

Answer: never. This is Theorem 1.38, and it leans on Theorem 1.31 (the counting theorem from Section 1.4) applied *twice*. Since basis B spans and C is independent, $|C| \leq |B|$. But by the same logic with roles swapped, $|B| \leq |C|$. So $|B| = |C|$ — forced.

This is exactly why **dimension** is a legitimate, well-defined number attached to a vector space: $\dim(V)$ is the size of *any* basis, and it doesn't matter which one you pick to count.

5.6 Key dimension facts

- $\dim(F^n) = n$, standard basis $\{e_1, \ldots, e_n\}$.
- $\dim(P_n(F)) = n + 1$ — careful, it's $n + 1$, not n! P_n includes degrees 0 through n, so the basis $\{1, t, t^2, \ldots, t^n\}$ has $n + 1$ elements.
- $\dim(\{0\}) = 0$ (the basis is $\emptyset$).
- $\dim(P(F)) = \infty$ — no finite basis exists, since $P(F)$ contains polynomials of arbitrarily high degree. By Theorem 1.40(3): infinite-dimensional $\iff$ there's an infinite linearly independent subset, e.g. $\{1, t, t^2, t^3, \ldots\}$.
- **Field matters for dimension.** $\mathbb{C}$ as a *complex* vector space has dim $= 1$ (basis $\{1\}$), but as a *real* vector space, dim $= 2$ (basis $\{1, i\}$) — since over $\mathbb{R}$ you need two independent "directions" (1 and i) to reach any complex number $a + bi$.

Worked Example s: computing a few dimensions

- $P_2(\mathbb{R}) = \{a + bt + ct^2\}$ has basis $\{1, t, t^2\}$, so dim $= 3$ (not 2!).
- $\mathbb{R}^{2\times 2}$ has basis $\{\left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right), \left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right), \left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right), \left(\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right)\}$, so dim $= 4$.
- The plane $U = \{(a, b, c) : a + b + c = 0\} \subset \mathbb{R}^3$ (from our Section 1.2 example) has dim $= 2$: a basis is $\{(1, -1, 0), (1, 0, -1)\}$ (check: both satisfy $a + b + c = 0$, and they're independent since neither is a multiple of the other).

5.7 Practical shortcut (Theorem 1.41) — saves you real work

If you already know $\dim(V) = n$, you only need to check *one* of the two basis conditions, not both, for a candidate set B with exactly n vectors:

- $|B| = n$ **and** B spans $V \implies$ automatically a basis (independence is free).
- $|B| = n$ **and** B is independent $\implies$ automatically a basis (spanning is free).

Worked Example : putting it all together

Is $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ a basis of $\mathbb{R}^3$?
We know $\dim(\mathbb{R}^3) = 3$, and we have exactly 3 vectors — so by Theorem 1.41 we only need to check **one** property. Check independence:

$$a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 1) = (0, 0, 0) \implies (a + c,\ a + b,\ b + c) = (0, 0, 0).$$

From these: $a = -c$; then $a+b = 0 \Rightarrow b = -a = c$; then $b+c = 0 \Rightarrow c = -b = -c \Rightarrow c = 0$; then $b = 0,\ a = 0$. Only the trivial solution — **independent**. Since we have exactly $\dim(\mathbb{R}^3) = 3$ vectors and independence holds, this is **automatically a basis**. No separate check of spanning needed.

Key Takeaway

Basis = spanning + independent = minimal spanning set = maximal independent set (all equivalent). Every basis of a given finite-dimensional space has the *same* number of vectors

— that number is the dimension. Once you know the dimension, checking a candidate basis only requires verifying *one* of the two properties, not both.

How the Five Sections Fit Together

1.1 Vector Space	the algebraic playground (8 axioms)
$\downarrow$	
1.2 Subspaces	smaller playgrounds living inside the big one
$\downarrow$	
1.3 Linear Span	smallest subspace generated by a set of vectors
$\downarrow$	
1.4 Linear Independence	"does this set have redundancy?"
$\downarrow$	
1.5 Basis = Span + Independence	the most efficient generating set dimension = size of that set (well-defined!)

Diagnosing which section a problem is testing

- "Show this set is/isn't a vector space" $\to$ **1.1**: check all 8 axioms, or find one that fails.
- "Show this subset is/isn't a subspace" $\to$ **1.2**: closure test — contains 0, closed under + and scalar mult.
- "What's the span of...?" / "Does this set span...?" $\to$ **1.3**: write the general linear combination, see what it can produce.
- "Are these vectors dependent/independent?" $\to$ **1.4**: set the linear combination equal to zero, solve for coefficients.
- "Find a basis / the dimension of..." $\to$ **1.5**: find a spanning set, trim redundant vectors until independent (or verify count against known dimension).

Common Points of Confusion — Reread If Something Didn't Click

1. **"Vector" is an overloaded word.** A "vector" here might be a tuple, a matrix, a polynomial, or a function. Don't let the word trick you into picturing only arrows — always ask "vector *in what space*?"

2. **Span vs. subspace vs. spanning set** are related but distinct: *span* is an operation (takes a set, produces a subspace); a *subspace* is a type of set; a *spanning set* is any set whose span equals a given space. $\mathrm{span}(S)$ is always a subspace, but S itself usually is not.

3. **Independent $\neq$ "no vector is a multiple of another."** That shortcut only works for pairs. For 3+ vectors, always fall back to the full definition.

4. **A basis is not unique, but the dimension is.** $\mathbb{R}^3$ has infinitely many valid bases (the standard one is just the most convenient), but every single one has exactly 3 vectors.

5. **"Degree at most n" polynomials have dimension $n + 1$, not n** — the +1 comes from including the constant term (degree 0).

If a specific exercise from the Section 1.1–1.5 problem sets is still giving you trouble, ask and we'll walk through it together, step by step.